

QoI-Preserving Lossy Compression for SAM3D Turbulence Data

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Abstract

We have applied Quantity-of-Interest (QoI) preserving lossy compression to SAM3D Large-Eddy Simulation (LES) data from the LASSO-ENA project. Given 6 initial 3D data arrays and 9 derived QoI data arrays, we bound all 15 arrays with sepecified relative error bound only by compressing the 6 initial arrays with certain error bound. The 9 QoIs we consider include 5 bilinear QOIs, 3 quadratic variances, and 1 higher-order skewness. Since QPET fails when one initial data point is used to derive multiple QoI data points and it does not support bounding multiple QoIs, we only adopted its method to calculate point-wise error bound when compressing initial data arrays. While we successfully bound all initial arrays and QoI arrays, the overall compression ratio is relatively low(8.5x) even we used a loose error bound(1e-2). We further introduce an empirical per-field binary-search framework supporting SZ3, zfp, and SPERR that substantially improves the compression ratio (up to 30.5 $\times$ with SZ3 at $\tau = 10^{-2}$) while still bounding all nine QoIs.

1 Data Overview

The data originates from the LASSO-ENA (Large-Eddy Simulation ARM Symbiotic Simulation and Observation – Eastern North Atlantic) project, using the SAM (System for Atmospheric Modeling) model v6.10.3 with LASSO modifications.

Six 3D variables are used for compression, each with shape $(N_z, N_y, N_x) = (260, 256, 256)$ and dtype `float32`:

Table 1: Primary 3D Variables for Compression

Variable	Description	Units	Typical Range
U	Zonal wind component	m/s	3 – 19
V	Meridional wind component	m/s	−22 – −3
W	Vertical wind component	m/s	−0.27 – 0.31
PP	Pressure perturbation	Pa	−0.72 – 0.84
$TABS$	Absolute temperature	K	240 – 287
QV	Water vapor mixing ratio	g/kg	0.02 – 7.9

Total timesteps: 96

Dimension per array: $(N_z, N_y, N_x) = (260, 256, 256)$

Memory footprint: 68.2 MB per variable (float32), 408.9 MB per timestep for 6 variables, 38.4 GB total for 96 timesteps

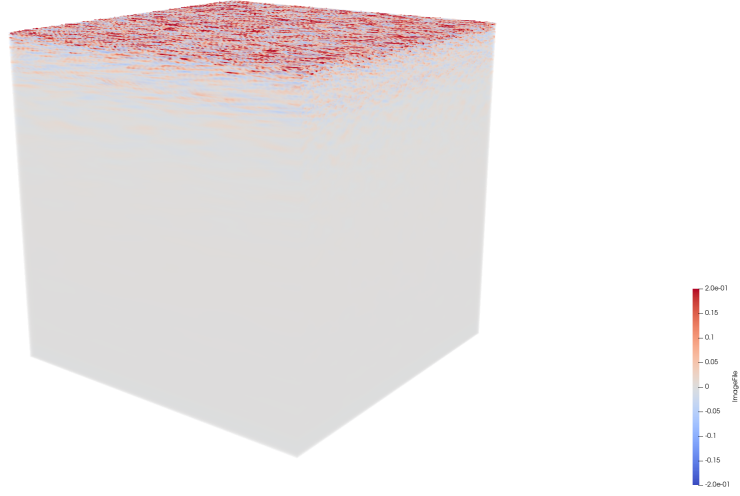


Figure 1: Pressure perturbation of time step 1 (N_z, N_y, N_x) = (260, 256, 256)

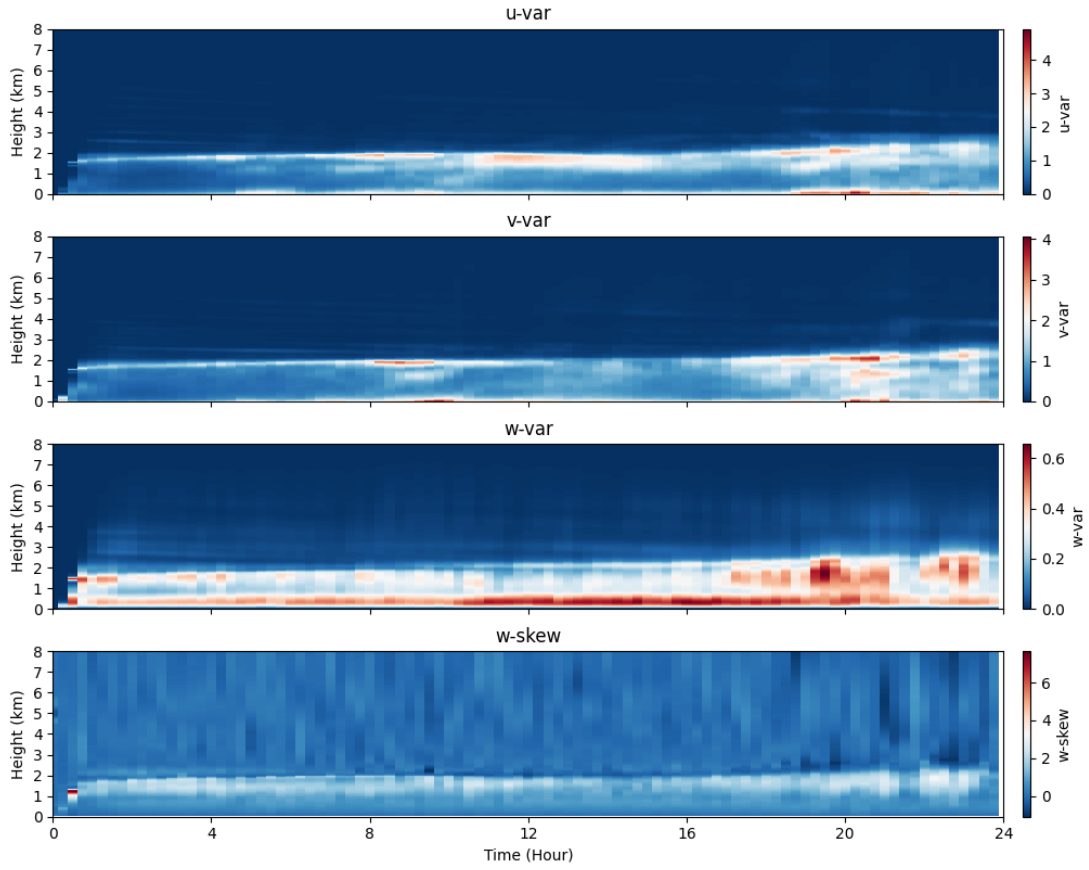


Figure 2: Turbulence statistics derived from 96 timesteps: velocity variances (uvar, vvar, wvar) and vertical velocity skewness (wskew)

2 Quantities of Interest (QoIs)

2.1 Mathematical Notations

Table 2: Symbol Definitions

Symbol	Description
ϕ	Generic field variable (e.g., u, v, w, T, p, q_v)
$\overline{\phi}(z)$	Horizontal mean of ϕ at height z
ϕ'	Turbulent perturbation ($\phi' = \phi - \overline{\phi}$)
$\langle \cdot \rangle_{xy}$	Horizontal averaging operator over x and y
N_x, N_y, N_z	Number of grid points in x, y , and z directions
$\sigma_w(z)$	Standard deviation of vertical velocity at height z
u', v', w'	Perturbations of velocity components
T', p', q'_v	Perturbations of temperature, pressure, and water vapor

All derived QoIs are based on Reynolds decomposition, separating each field ϕ into a horizontal mean and turbulent perturbation:

$$\phi(x, y, z) = \overline{\phi}(z) + \phi'(x, y, z) \quad (1)$$

where the overbar denotes horizontal averaging at each height level:

$$\overline{\phi}(z) = \langle \phi \rangle_{xy} = \frac{1}{N_x N_y} \sum_{i,j} \phi(x_i, y_j, z) \quad (2)$$

The perturbation is computed as:

$$\phi' = \phi - \langle \phi \rangle_{xy} \quad (3)$$

2.2 Zonal Momentum Flux – wpup (Bilinear)

$$\overline{w'u'}(z) = \langle w'(x, y, z) \cdot u'(x, y, z) \rangle_{xy} \quad (4)$$

2.3 Meridional Momentum Flux – wpvp (Bilinear)

$$\overline{w'v'}(z) = \langle w'(x, y, z) \cdot v'(x, y, z) \rangle_{xy} \quad (5)$$

2.4 Pressure-Velocity Covariance – wppp (Bilinear)

$$\overline{w'p'}(z) = \langle w'(x, y, z) \cdot p'(x, y, z) \rangle_{xy} \quad (6)$$

2.5 Sensible Heat Flux – wptp (Bilinear)

$$\overline{w'T'}(z) = \langle w'(x, y, z) \cdot T'(x, y, z) \rangle_{xy} \quad (7)$$

2.6 Moisture Flux – wpqvp (Bilinear)

$$\overline{w'q'_v}(z) = \langle w'(x, y, z) \cdot q'_v(x, y, z) \rangle_{xy} \quad (8)$$

2.7 Zonal Velocity Variance – uvar (Quadratic)

$$\overline{u'^2}(z) = \langle u'(x, y, z)^2 \rangle_{xy} \quad (9)$$

2.8 Meridional Velocity Variance – vvar (Quadratic)

$$\overline{v'^2}(z) = \langle v'(x, y, z)^2 \rangle_{xy} \quad (10)$$

2.9 Vertical Velocity Variance – wvar (Quadratic)

$$\overline{w'^2}(z) = \langle w'(x, y, z)^2 \rangle_{xy} = \sigma_w^2(z) \quad (11)$$

2.10 Vertical Velocity Skewness – wskew (Higher-Order)

$$S_w(z) = \left\langle \left(\frac{w'}{\sigma_w} \right)^3 \right\rangle_{xy} = \frac{\langle w'^3 \rangle_{xy}}{\sigma_w^3(z)} \quad (12)$$

where $\sigma_w(z) = \sqrt{\overline{w'^2}(z)}$ is the standard deviation of vertical velocity at height z .

2.11 Summary of QoIs

Table 3: Summary of Quantities of Interest

QoI	Formula	Type	Dependent Fields
wpup	$\langle w'u' \rangle_{xy}$	Bilinear	W, U
wpvp	$\langle w'v' \rangle_{xy}$	Bilinear	W, V
wppp	$\langle w'p' \rangle_{xy}$	Bilinear	W, PP
wptp	$\langle w'T' \rangle_{xy}$	Bilinear	$W, TABS$
wpqvp	$\langle w'q'_v \rangle_{xy}$	Bilinear	W, QV
uvar	$\langle u'^2 \rangle_{xy}$	Quadratic	U
vvar	$\langle v'^2 \rangle_{xy}$	Quadratic	V
wvar	$\langle w'^2 \rangle_{xy}$	Quadratic	W
wskew	$\langle w'^3 \rangle_{xy} / \sigma_w^3$	Cubic/Rational	W

Note that the W field appears in all 9 QoIs, making it the most constrained field in QoI-preserving compression.

Takeaway: All QoIs are non-trivial functions because they are all non-linear, non operator-separable, and each single data point in the initial data array contributes to multiple QoIs.

3 Method

3.1 Input, Output, and Error Bounds

Input files: Six 3D binary arrays stored as raw float32 files:

- `U_260x256x256.raw`, `V_260x256x256.raw`, `W_260x256x256.raw`
- `PP_260x256x256.raw`, `TABS_260x256x256.raw`, `QV_260x256x256.raw`

Input error bounds: User specifies relative error bounds for:

- 6 field arrays: $\epsilon_U, \epsilon_V, \epsilon_W, \epsilon_{PP}, \epsilon_{TABS}, \epsilon_{QV}$
- 9 QoI arrays: $\tau_{wpup}, \tau_{wpvp}, \tau_{wppp}, \tau_{wptp}, \tau_{wpqvp}, \tau_{uvar}, \tau_{vvar}, \tau_{wvar}, \tau_{wskew}$

Output files: Six SZ3-compressed files with calculated absolute error bounds that satisfy both field and QoI constraints.

3.2 Point-wise Error Bound Derivation

Following the QPET framework, for a multivariate QoI function $F(x_1, \dots, x_n)$ that is not in variable-separated format, we use its first-order Taylor approximation:

$$F(x'_1, \dots, x'_n) \approx F(x_1, \dots, x_n) + \sum_i \frac{\partial F}{\partial x_i}(x_i)(x'_i - x_i) \quad (13)$$

This reduces the multivariate QoI to a linear combination format $F = C + \sum_i \alpha_i f(x_i)$ where $\alpha_i = \frac{\partial F}{\partial x_i}$ and $f(x) = x$. The point-wise error bound for each data point is computed using:

$$t = \frac{T}{\sum_i |\alpha_i|} \quad (14)$$

where T is the QoI error threshold. This deterministic bound guarantees that if each point-wise error $|D_i| \leq t$, then $|\sum_i \alpha_i D_i| \leq T$.

Takeaway: Since each single data point in the initial data array contributes to multiple QoIs, the simple edition approach in QPET (which corrects outliers after compression) is not applicable. Instead, we must hard-bound all data points during compression.

3.3 Sequential Bound Calculation for Multiple QoIs

When bounding multiple QoIs that share common input fields, the **sequence of error bound calculation matters**. Consider our 9 QoIs where the W field appears in all of them:

Problem: The skewness QoI depends on $\sigma_w = \sqrt{\text{wvar}}$, so the wvar error must be known before computing the wskew bound. Similarly, bilinear QoIs like $\langle w'u' \rangle$ need the W bound to allocate the remaining error budget to other fields.

Our approach:

1. **Step 1: Compute wvar bound first.** Since $\text{wvar} = \langle w'^2 \rangle_{xy}$ is quadratic in W , we derive:

$$\epsilon_W^{\text{wvar}} = \frac{\tau_{\text{wvar}} \cdot \text{range}(\text{wvar})}{4 \cdot \max_z \langle |w'| \rangle} \quad (15)$$

2. **Step 2: Compute wskew bound using known wvar error.** The skewness $S_w = \langle w'^3 \rangle / \sigma_w^3$ has per-datapoint partial derivatives:

$$\frac{\partial f_i}{\partial w'_i} = \frac{3(w'_i)^2}{\sigma^3}, \quad \frac{\partial f_i}{\partial (\sigma^2)} = -\frac{3(w'_i)^3}{2\sigma^5} \quad (16)$$

The σ^2 term uses the known wvar error from Step 1.

3. **Step 3: Compute bilinear QoI bounds.** For $\langle w'\phi' \rangle$, with the W bound already determined, we allocate the remaining error budget to the other field ϕ :

$$\epsilon_\phi = \frac{\tau \cdot \text{range}(\text{QoI}) - 2\epsilon_W \cdot \max_z \langle |\phi'| \rangle}{2 \cdot \max_z \langle |w'| \rangle} \quad (17)$$

The final field error bound is the **minimum** across all QoI-derived bounds:

$$\epsilon_W = \min(\epsilon_W^{\text{user}}, \epsilon_W^{\text{wvar}}, \epsilon_W^{\text{wskew}}, \epsilon_W^{\text{wpup}}, \epsilon_W^{\text{wpvp}}, \epsilon_W^{\text{wppp}}, \epsilon_W^{\text{wptp}}, \epsilon_W^{\text{wpqvp}}) \quad (18)$$

Takeaway: When one initial data array is used in multiple QoIs, the sequence of processing QoIs to derive error bounds for the initial data array is crucial to the overall compression ratio.

4 Compression Results

We evaluate QoI-preserving compression at three relative error bounds: 10^{-2} , 10^{-3} , and 10^{-4} .

4.1 Relative Error Bound = 10^{-2}

Table 4: Compression Results ($\tau = 10^{-2}$)

Field	Calculated Abs. Bound	Calculated Rel. Bound	Compression Ratio
U	6.17×10^{-4}	3.92×10^{-5}	24.32x
V	1.86×10^{-3}	1.00×10^{-4}	30.09x
W	1.93×10^{-9}	3.33×10^{-9}	1.89x
PP	2.30×10^{-4}	1.47×10^{-4}	30.29x
$TABS$	1.35×10^{-4}	2.86×10^{-6}	25.13x
QV	6.50×10^{-4}	8.21×10^{-5}	35.46x
Overall	–	–	8.50x

4.2 Relative Error Bound = 10^{-3}

Table 5: Compression Results ($\tau = 10^{-3}$)

Field	Calculated Abs. Bound	Calculated Rel. Bound	Compression Ratio
U	6.17×10^{-5}	3.92×10^{-6}	11.26x
V	1.86×10^{-4}	1.00×10^{-5}	13.87x
W	1.95×10^{-10}	3.36×10^{-10}	1.37x
PP	2.30×10^{-5}	1.47×10^{-5}	10.68x
$TABS$	1.35×10^{-5}	2.86×10^{-7}	10.02x
QV	6.50×10^{-5}	8.21×10^{-6}	17.81x
Overall	–	–	5.26x

4.3 Relative Error Bound = 10^{-4}

Table 6: Compression Results ($\tau = 10^{-4}$)

Field	Calculated Abs. Bound	Calculated Rel. Bound	Compression Ratio
U	6.17×10^{-6}	3.92×10^{-7}	4.83x
V	1.86×10^{-5}	1.00×10^{-6}	6.09x
W	1.94×10^{-11}	3.36×10^{-11}	1.15x
PP	2.30×10^{-6}	1.47×10^{-6}	4.85x
$TABS$	1.35×10^{-6}	2.86×10^{-8}	10.20x
QV	6.50×10^{-6}	8.21×10^{-7}	9.18x
Overall	–	–	3.63x

4.4 Summary

Table 7: Overall Compression Ratio vs. Relative Error Bound

Relative Error Bound	Overall Compression Ratio
10^{-2}	8.50x
10^{-3}	5.26x
10^{-4}	3.63x

The W field is the bottleneck, requiring extremely tight absolute error bounds ($\sim 10^{-9}$ to 10^{-11}) to satisfy the w_{skew} and w_{var} QoI constraints. This results in minimal compression for W (1.15x–1.89x), which limits the overall compression ratio.

Takeaway: The W data array (z-velocity) is over-bounded because of the complexity of w_{skewness} , which requires a better method to get the error back-propagated to the original data array.

4.5 QoI Comparison

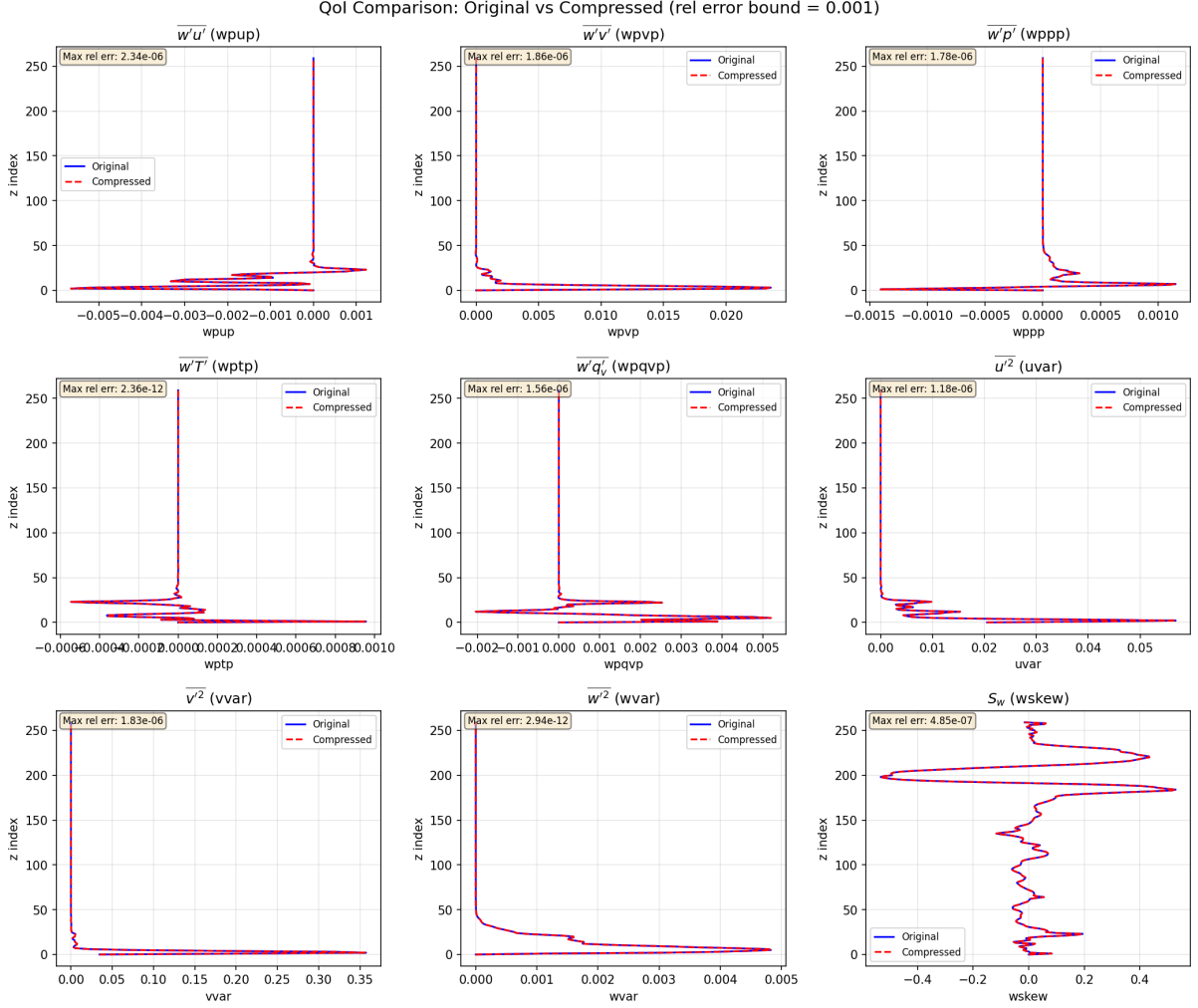


Figure 3: Comparison of 9 QoIs computed from original data vs. compressed/decompressed data (relative error bound = 10^{-3}). Each QoI is a 1D profile of length $N_z = 260$. Blue: original, Red dashed: compressed. All QoIs from original data and compressed data are almost identical, which indicates over conservative error bounds

5 Empirical Per-Field Bound Search

The analytical QPET bounds derived above are conservative: the first-order Taylor propagation over-bounds W (driving ϵ_W to $\sim 10^{-9}$) and limits the overall ratio. We therefore introduce an *empirical* per-field search that finds, for each field, the loosest absolute error bound that still keeps every dependent QoI within its target, by directly compressing, decompressing, and measuring the resulting QoI error. The framework is compressor-agnostic and is evaluated with three error-bounded compressors: SZ3, zfp, and SPERR.

5.1 QoI Error Metric

For each QoI we use the range-relative maximum error over the vertical profile:

$$E(\text{QoI}) = \frac{\max_z |\text{QoI}_{\text{orig}}(z) - \text{QoI}_{\text{dec}}(z)|}{\max_z \text{QoI}_{\text{orig}}(z) - \min_z \text{QoI}_{\text{orig}}(z)} \leq \tau, \quad (19)$$

where τ is the target relative QoI bound, applied uniformly to all nine QoIs.

5.2 Per-Field Binary Search

For a single field, the highest admissible bound is found by **binary search** over the absolute error bound ϵ in logarithmic space on $[r \cdot 10^{-7}, r \cdot 0.5]$, where r is the field’s value range. At each step the field is compressed and decompressed at the trial ϵ , the dependent QoI(s) are recomputed, and the trial passes iff all satisfy $E \leq \tau$. If the midpoint passes, the floor is raised (looser bound); otherwise the ceiling is lowered (tighter bound). The largest passing bound, scaled by a safety margin (0.9), is returned. The iteration count is capped to limit the impact of compressor non-monotonicity.

5.3 QoI Search Sequence

A key property is **separability**: because each perturbation subtracts a per-field horizontal mean, a QoI depends only on its own field(s); compressing field A never alters a QoI that does not depend on A . A per-field search with the coupled field(s) fixed therefore reproduces the all-fields result exactly. Since W appears in all nine QoIs but is constrained by single-field QoIs only ($wvar$, $wskew$), it is locked first:

1. **Phase 1 (single-field QoIs).** Search U via $wvar$, V via $vvar$, and W via both $wvar$ and $wskew$ (the tighter bound wins). Fix W at its decompressed array for the next phase.
2. **Phase 2 (multi-field QoIs, W fixed).** Search PP via $wppp$, $TABS$ via $wptp$, and QV via $wpgvp$; re-search U via wup and V via wvp . Each field’s final bound is the minimum of its Phase-1 and Phase-2 results.

This is the same dependency-ordered sequence motivated in the analytical method, but the bounds are determined empirically rather than via Taylor formulas.

5.4 W -Coupling Fallback

If a bilinear QoI cannot be met even with its partner field near-lossless — i.e. W ’s own error already exhausts the QoI budget — W is tightened and Phase 2 is restarted. Tightening W only improves every other QoI, so the step is always safe.

5.5 Final Verification and Tightening

After the per-field bounds are fixed, all six fields are compressed simultaneously and all nine QoIs are recomputed. Should any QoI marginally exceed its target — possible because of compressor non-monotonicity and the shared W coupling — the bounds of the fields feeding that QoI are **tightened** by a factor of 0.9 and the verification is repeated, for a small bounded number of rounds.

5.6 Lossless Fallback

A QoI that remains infeasible even with its fields near-lossless triggers **lossless** storage of those fields. In practice this affects only W , driven by $wskew$: the normalized third moment is hypersensitive to error at low-variance heights, so some compressors cannot bound it at any lossy setting. Lossless storage uses each backend’s native mode (zfp reversible, SZ3 with error bound 0) and falls back to raw storage where there is none (SPERR). Such fields are reported with bound 0.

6 Multi-Compressor Results

We run nine configurations: three compressors (SZ3, zfp, SPERR) at three relative QoI bounds (10^{-2} , 10^{-3} , 10^{-4}), each applied uniformly to all nine QoIs, on a single timestep (six `float32` fields, 408.9MB total). Every configuration satisfies all nine QoIs, using the lossless fallback where noted. Table 8 reports the overall ratio, Figure 4 and Table 9 the per-field ratio, Table 10 the per-field error bounds, and Table 11 the full per-configuration detail.

Table 8: Overall compression ratio for all nine configurations.

Compressor	$\tau = 10^{-2}$	$\tau = 10^{-3}$	$\tau = 10^{-4}$
SZ3	30.52x	18.30x	10.87x
zfp	6.23x	5.68x	5.09x
SPERR	15.80x	13.47x	5.13x

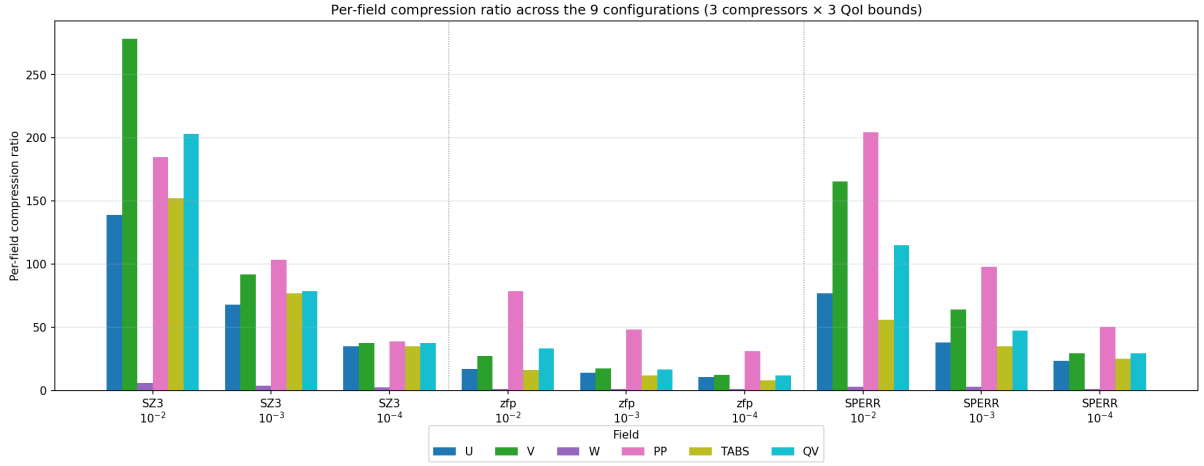


Figure 4: Per-field compression ratio for the nine configurations. Each cluster is one configuration (compressor at a given QoI bound) and contains one column per field ($U, V, W, PP, TABS, QV$). Dotted lines separate the three compressors.

Table 9: Per-field compression ratio across all nine configurations.

Field	SZ3			zfp			SPERR		
	10^{-2}	10^{-3}	10^{-4}	10^{-2}	10^{-3}	10^{-4}	10^{-2}	10^{-3}	10^{-4}
U	138.89x	68.11x	35.10x	17.16x	14.23x	10.49x	76.73x	37.88x	23.38x
V	278.43x	92.07x	37.42x	27.18x	17.45x	12.30x	165.33x	64.07x	29.24x
W	5.92x	3.75x	2.41x	1.31x	1.31x	1.31x	3.04x	2.91x	1.00x
PP	184.59x	103.36x	38.78x	78.70x	48.36x	30.97x	204.24x	97.92x	50.51x
TABS	152.21x	77.01x	34.85x	16.18x	11.90x	8.20x	56.15x	35.00x	25.00x
QV	203.23x	78.57x	37.35x	33.19x	16.41x	11.72x	114.86x	47.21x	29.48x
Overall	30.52x	18.30x	10.87x	6.23x	5.68x	5.09x	15.80x	13.47x	5.13x

Table 10: Per-field relative error bound found by the search across all nine configurations. “lossless denotes a field stored without loss.

Field	SZ3			zfp			SPERR		
	10^{-2}	10^{-3}	10^{-4}	10^{-2}	10^{-3}	10^{-4}	10^{-2}	10^{-3}	10^{-4}
U	1.61×10^{-3}	4.97×10^{-4}	1.14×10^{-4}	2.86×10^{-2}	1.43×10^{-2}	3.57×10^{-3}	3.19×10^{-3}	7.39×10^{-4}	1.54×10^{-4}
V	3.70×10^{-3}	1.04×10^{-3}	2.07×10^{-4}	9.68×10^{-2}	2.42×10^{-2}	6.05×10^{-3}	5.82×10^{-3}	1.67×10^{-3}	2.61×10^{-4}
W	4.13×10^{-6}	5.85×10^{-7}	6.56×10^{-8}	lossless	lossless	lossless	9.00×10^{-8}	6.56×10^{-8}	lossless
PP	4.90×10^{-3}	1.99×10^{-3}	2.97×10^{-4}	1.44×10^{-1}	3.61×10^{-2}	9.02×10^{-3}	6.11×10^{-3}	1.48×10^{-3}	2.93×10^{-4}
TABS	8.65×10^{-5}	3.12×10^{-5}	6.89×10^{-6}	4.77×10^{-3}	1.19×10^{-3}	1.49×10^{-4}	1.44×10^{-4}	4.23×10^{-5}	1.30×10^{-5}
QV	2.54×10^{-3}	4.78×10^{-4}	8.92×10^{-5}	1.14×10^{-1}	1.42×10^{-2}	3.56×10^{-3}	4.01×10^{-3}	7.15×10^{-4}	2.00×10^{-4}

Table 11: Full per-configuration detail: per-field absolute bound, relative bound, and compression ratio for every field in all nine configurations. Fields stored losslessly are shown with bound 0.

Field	Abs. bound	Rel. bound	Ratio
SZ3, $\tau = 10^{-2}$ — overall 30.52x			
U	2.542×10^{-2}	1.614×10^{-3}	138.89x
V	6.878×10^{-2}	3.698×10^{-3}	278.43x
W	2.391×10^{-6}	4.129×10^{-6}	5.92x
PP	7.642×10^{-3}	4.899×10^{-3}	184.59x
TABS	4.081×10^{-3}	8.646×10^{-5}	152.21x
QV	2.006×10^{-2}	2.536×10^{-3}	203.23x
SZ3, $\tau = 10^{-3}$ — overall 18.30x			
U	7.824×10^{-3}	4.967×10^{-4}	68.11x
V	1.937×10^{-2}	1.042×10^{-3}	92.07x
W	3.387×10^{-7}	5.850×10^{-7}	3.75x
PP	3.100×10^{-3}	1.987×10^{-3}	103.36x
TABS	1.475×10^{-3}	3.124×10^{-5}	77.01x
QV	3.783×10^{-3}	4.783×10^{-4}	78.57x
SZ3, $\tau = 10^{-4}$ — overall 10.87x			
U	1.793×10^{-3}	1.138×10^{-4}	35.10x
V	3.846×10^{-3}	2.068×10^{-4}	37.42x
W	3.799×10^{-8}	6.561×10^{-8}	2.41x
PP	4.633×10^{-4}	2.970×10^{-4}	38.78x
TABS	3.252×10^{-4}	6.890×10^{-6}	34.85x
QV	7.057×10^{-4}	8.922×10^{-5}	37.35x
zfp, $\tau = 10^{-2}$ — overall 6.23x			
U	4.500×10^{-1}	2.857×10^{-2}	17.16x
V	1.800	9.679×10^{-2}	27.18x
W	0	0	1.31x
PP	2.250×10^{-1}	1.442×10^{-1}	78.70x
TABS	2.250×10^{-1}	4.766×10^{-3}	16.18x
QV	9.000×10^{-1}	1.138×10^{-1}	33.19x
zfp, $\tau = 10^{-3}$ — overall 5.68x			
U	2.250×10^{-1}	1.428×10^{-2}	14.23x
V	4.500×10^{-1}	2.420×10^{-2}	17.45x
W	0	0	1.31x
PP	5.625×10^{-2}	3.606×10^{-2}	48.36x
TABS	5.625×10^{-2}	1.192×10^{-3}	11.90x
QV	1.125×10^{-1}	1.422×10^{-2}	16.41x
zfp, $\tau = 10^{-4}$ — overall 5.09x			
U	5.625×10^{-2}	3.571×10^{-3}	10.49x
V	1.125×10^{-1}	6.049×10^{-3}	12.30x

(continued)

Field	Abs. bound	Rel. bound	Ratio
W	0	0	1.31x
PP	1.406×10^{-2}	9.015×10^{-3}	30.97x
TABS	7.031×10^{-3}	1.490×10^{-4}	8.20x
QV	2.812×10^{-2}	3.556×10^{-3}	11.72x
SPERR, $\tau = 10^{-2}$ — overall 15.80x			
U	5.032×10^{-2}	3.194×10^{-3}	76.73x
V	1.082×10^{-1}	5.820×10^{-3}	165.33x
W	5.212×10^{-8}	9.000×10^{-8}	3.04x
PP	9.536×10^{-3}	6.114×10^{-3}	204.24x
TABS	6.800×10^{-3}	1.441×10^{-4}	56.15x
QV	3.171×10^{-2}	4.009×10^{-3}	114.86x
SPERR, $\tau = 10^{-3}$ — overall 13.47x			
U	1.164×10^{-2}	7.392×10^{-4}	37.88x
V	3.111×10^{-2}	1.673×10^{-3}	64.07x
W	3.799×10^{-8}	6.561×10^{-8}	2.91x
PP	2.313×10^{-3}	1.483×10^{-3}	97.92x
TABS	1.997×10^{-3}	4.231×10^{-5}	35.00x
QV	5.658×10^{-3}	7.152×10^{-4}	47.21x
SPERR, $\tau = 10^{-4}$ — overall 5.13x			
U	2.434×10^{-3}	1.545×10^{-4}	23.38x
V	4.858×10^{-3}	2.612×10^{-4}	29.24x
W	0	0	1.00x
PP	4.574×10^{-4}	2.933×10^{-4}	50.51x
TABS	6.124×10^{-4}	1.297×10^{-5}	25.00x
QV	1.578×10^{-3}	1.995×10^{-4}	29.48x

Takeaway: SZ3 gives the best QoI-preserving compression at every bound ($30.5\times$ / $18.3\times$ / $10.9\times$) — about $2\times$ SPERR and $3\text{--}5\times$ zfp — and is the only compressor that preserves *wskew* without storing *W* losslessly. SPERR is competitive only at the looser bounds and collapses at 10^{-4} (raw *W*, as it has no lossless mode), while zfp is weakest throughout and must store *W* losslessly at every bound.

Takeaway: *W* is the universal bottleneck: the skewness *wskew* forces its error bound roughly three orders of magnitude tighter than any other field, and every lossless fallback in this study is *W*. Preserving vertical-velocity skewness is what caps the overall compression ratio.